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Domain Document

Business Process & Terminology — Airline Flight Delay Analysis

Data Source	Maven Analytics — Airline Flight Delays (U.S. DOT, 2015)
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1. Purpose

This document explains the airline industry business process behind the dataset and defines the domain terminology used throughout the project. It gives anyone reviewing the dashboard — even without an aviation background — enough context to interpret the metrics correctly.

2. The Flight Lifecycle (Business Process)

Every row in the flights table represents one scheduled flight operation, from the moment it is scheduled to its completion. The lifecycle has five stages:

1. **Scheduling** — the airline publishes a scheduled departure and arrival time for a given flight number, origin airport, and destination airport, along with a planned block/elapsed time.
2. **Departure** — the aircraft pushes back from the gate and takes off. Two sub-events are tracked: `DEPARTURE_TIME` (when the flight actually leaves the gate) and `WHEELS_OFF` (when it actually leaves the ground, after taxiing).
3. **En-route (Air Time)** — the time the aircraft spends actually airborne, from wheels-off to wheels-on, tracked as `AIR_TIME`.
4. **Arrival** — the aircraft lands (`WHEELS_ON`) and taxis to the gate, arriving at `ARRIVAL_TIME`.
5. **Disposition** — the flight is recorded as completed on-time/late, **cancelled** (never operated), or **diverted** (landed at an airport other than the scheduled destination).

A "delay" is simply the gap, in minutes, between the scheduled time and the actual time at the departure or arrival milestone. A flight can depart late but still arrive close to on-time if the crew makes up time in the air (this is common and is why departure delay and arrival delay are tracked separately).

3. Delay Cause Categories

When a flight's arrival delay is 15 minutes or more, the U.S. DOT requires airlines to attribute the delay minutes across up to five standardized cause categories. A single flight can have delay minutes split across more than one category. These are the categories used in this dataset:

Category	Column	Business Meaning	Controllable?
Air Carrier Delay	`AIRLINE_DELAY`	Delay caused by the airline itself — e.g. crew scheduling, aircraft cleaning, baggage handling, maintenance.	Yes
Late-Arriving Aircraft	`LATE_AIRCRAFT_DELAY`	The incoming aircraft for this flight arrived late from its previous leg, pushing this departure back (a "ripple effect").	Indirectly (network effect)
National Air System (NAS) Delay	`AIR_SYSTEM_DELAY`	Delay from air-traffic-control, airspace congestion, or non-extreme weather affecting the broader air system.	No
Weather Delay	`WEATHER_DELAY`	Delay caused directly by extreme weather at the origin/destination (severe storms, ice, etc.).	No
Security Delay	`SECURITY_DELAY`	Delay from security incidents such as evacuations, screening equipment failure, or long security lines.	No

Note: these five categories only apply to **delay minutes**. Cancellations use a separate, single-letter reason code described below.

4. Cancellation Reasons

If a flight never operated, it is flagged `CANCELLED = 1` and given exactly one reason code:

Code	Meaning
A	Airline / Carrier — operational decision by the airline (e.g. crew, mechanical, staffing).
B	Weather — severe weather at origin, destination, or en route.
C	National Air System — air-traffic control or airspace-driven cancellation.
D	Security — security-related cancellation.

5. Diversions

A **diverted** flight (`DIVERTED = 1`) is one that departed as planned but landed at an airport other than its scheduled destination, typically due to weather, a medical emergency, or a mechanical issue, before eventually continuing or terminating its journey. Diverted flights are operationally different from both on-time flights and cancellations and are tracked as their own status.

6. Key Domain Terms & Glossary

Term	Definition
IATA Code	A globally recognized 2-letter code for an airline (e.g. "AA" for American Airlines) or 3-letter code for an airport (e.g. "JFK"), issued by the International Air Transport Association.
Tail Number	The unique registration number painted on an aircraft's tail, identifying the specific physical airplane that operated the flight.
Block Time / Scheduled Time	The total planned duration of a flight from gate departure to gate arrival, as published in the airline's schedule.
Elapsed Time	The actual total gate-to-gate duration of the flight as it occurred, comparable to Scheduled Time.
Taxi-Out	Time spent moving from the gate to the runway before takeoff.
Taxi-In	Time spent moving from the runway to the gate after landing.
Air Time	Time spent actually airborne, between wheels-off and wheels-on.
On-Time Performance (OTP)	The industry-standard measure of the percentage of flights that depart or arrive within 15 minutes of schedule.
Origin / Destination Airport	The airport a flight departs from ("origin") and the airport it is scheduled to land at ("destination").
Route	A specific origin–destination airport pair (e.g. JFK → LAX), used for route-level performance analysis.
Carrier / Airline	The company operating the flight (e.g. Delta Air Lines, United Airlines).

7. Industry Context

In the United States, airlines with at least 0.5% of total domestic scheduled-service passenger revenue are required to report on-time performance data to the U.S. DOT, which publishes it in the monthly Air Travel Consumer Report. This dataset is a research-friendly compilation of that reporting for calendar year 2015, covering the major U.S. carriers operating at the time. Understanding this origin matters for two reasons: the data reflects *reported, regulated* operational statistics rather than raw airline internal systems data, and the metrics defined in the KPI Document (Section 5 of this project) intentionally follow the same on-time and cancellation definitions used in official DOT reporting, so dashboard results are comparable to publicly known industry benchmarks.

8. Why This Matters to the Business

Delay minutes and cancellation reasons are not just descriptive statistics — they map directly to business decisions. Weather and NAS delays are largely outside an airline's control and point to schedule padding or route/timing adjustments as the mitigation. Air Carrier and Late-Aircraft delays are within operational control and point to crew scheduling, ground-operations efficiency, and aircraft turnaround practices as areas for improvement. Separating these categories is the analytical foundation for every KPI and dashboard visual defined later in this document set.